Electrical Science
ENGR 2303

Video 4

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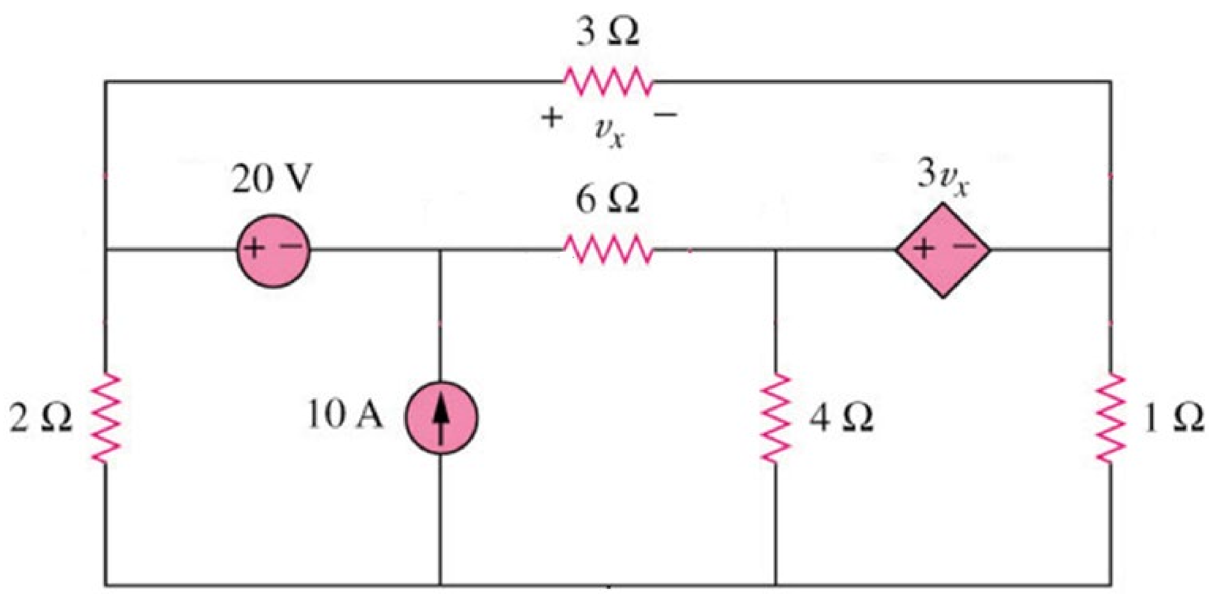
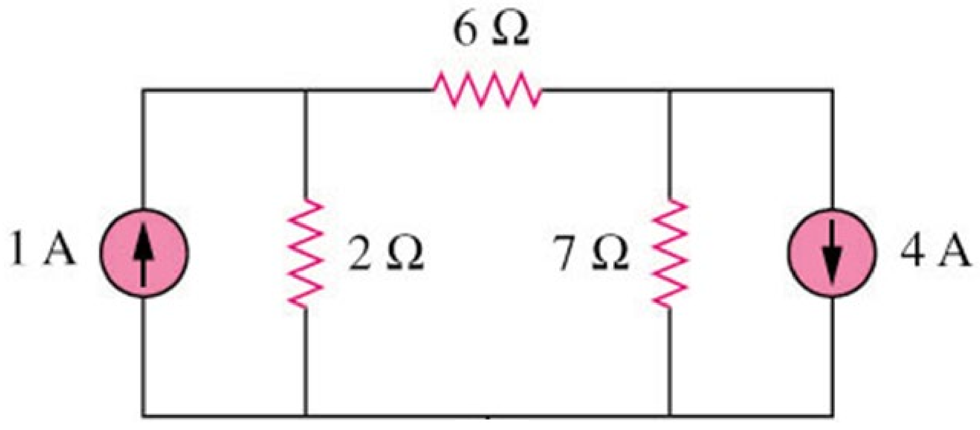
Nodal Analysis

- Nodal analysis.
- Nodal analysis with dependent sources.
- Nodal analysis with voltage sources.
- Examples



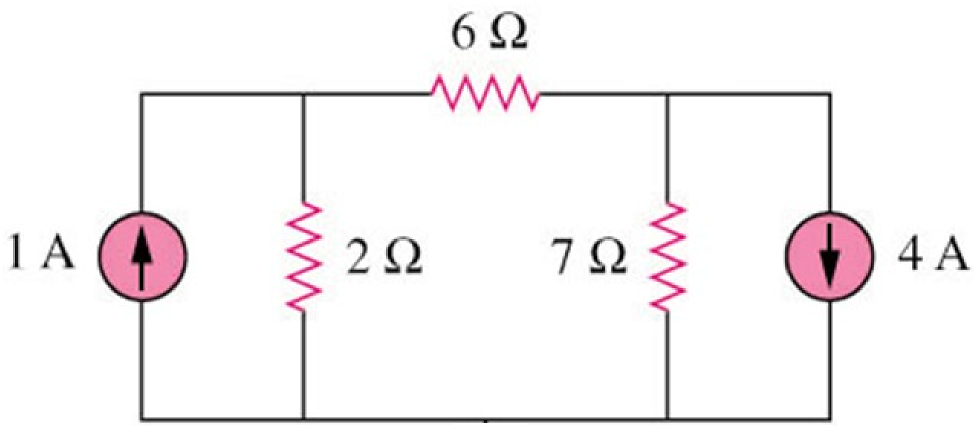
Nodal Analysis Steps

1. Select a node as the reference node.
2. Assign voltages $v_1, v_2, \ldots, v_{n-1}$ to the remaining n-1 nodes. The voltages are referenced with respect to the reference node.
3. Apply KCL to each of the n-1 non-reference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
4. Solve the resulting simultaneous equations to obtain the unknown node voltages.



Nodal Analysis Example

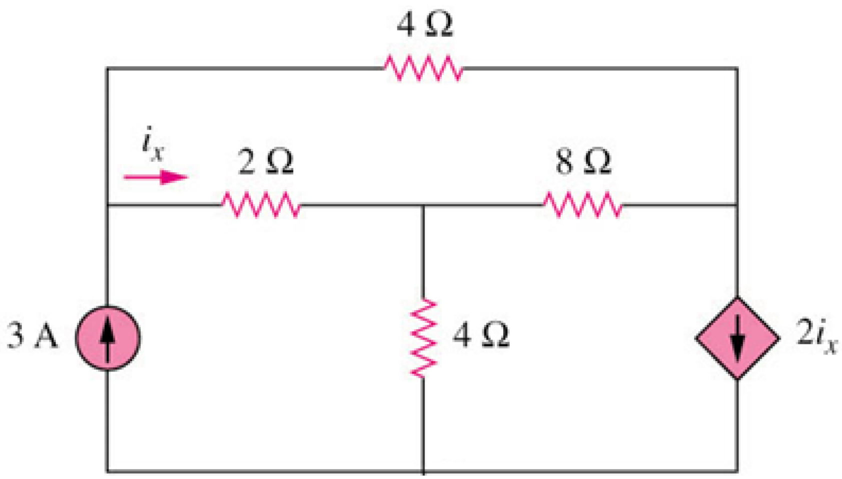
Find the voltages across the resistors.



Answer v_1 = -2V, v_2 = -14V

Nodal Analysis with Dependent source

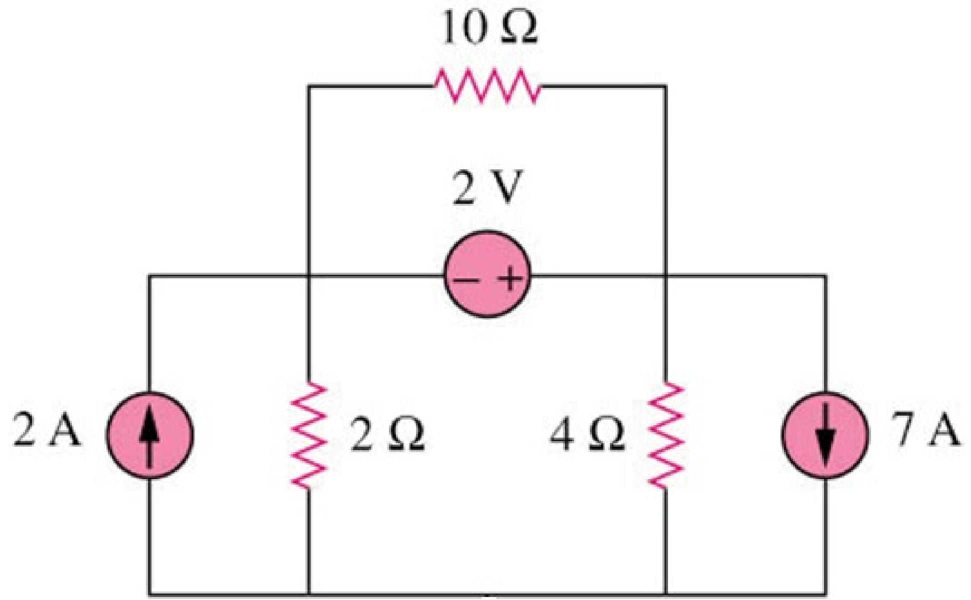
Find the voltages and currents across the resistors.



answer $v_1 = 4.8V$, $v_2 = 2.4V$, $v_3 = -2.4V$

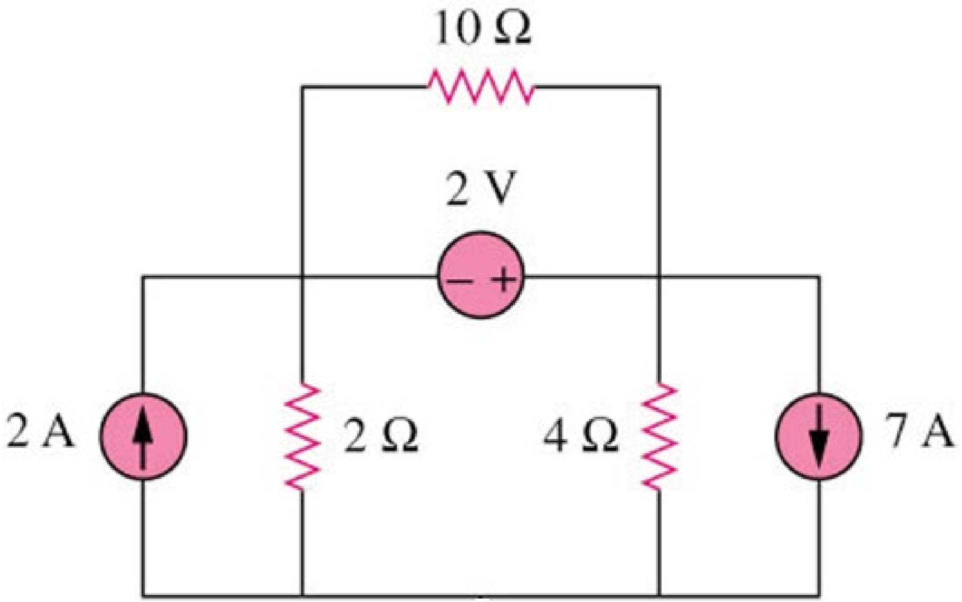
Nodal Analysis with Voltage Source

1. Form a super-node by enclosing a voltage source connected between two non-reference nodes and any elements connected in parallel with it.
2. Take off all voltage sources in super-nodes and apply KCL to super-nodes.
3. Put voltage sources back to the nodes and apply KVL to relative loops.



Nodal Analysis with Voltage Source

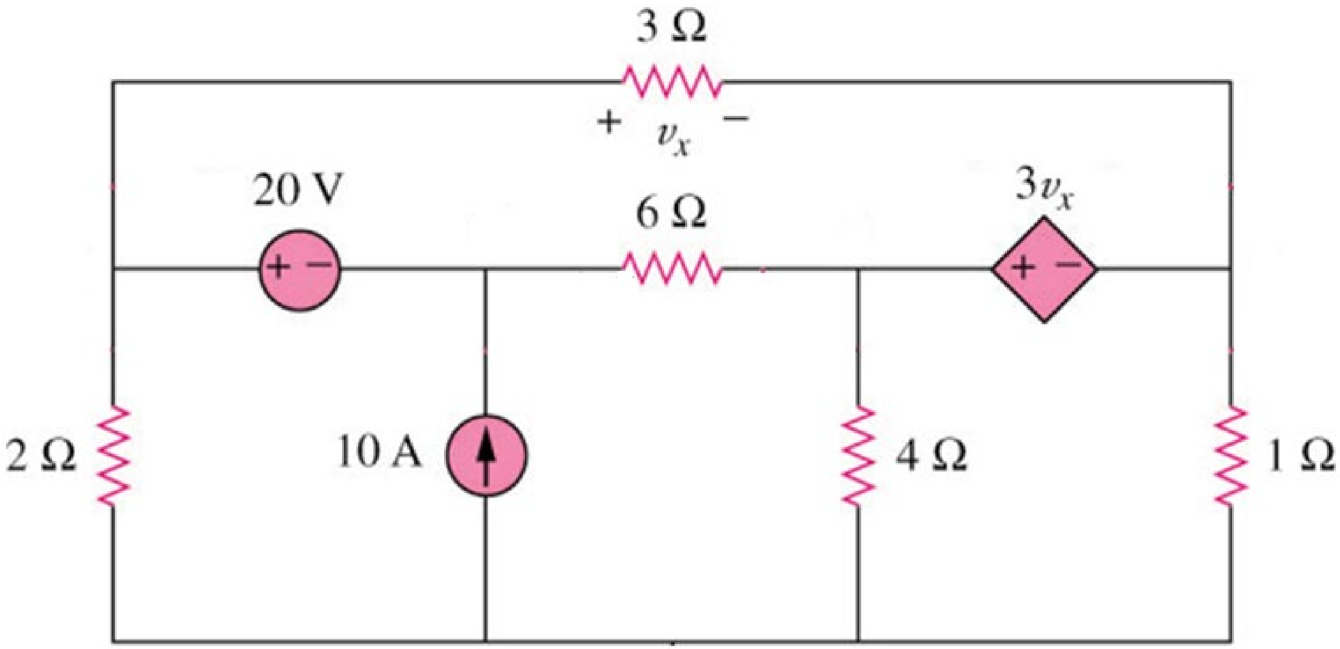
Find the voltages and currents at the resistors



answer v_1= -22/3 V, v_2 = -16/3 V

Nodal Analysis with Independent Voltage Source

Find the voltages at the nodes



$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 5 & 1 & -1 & -2 \\ 4 & 2 & -5 & -16 \\ -1 & 1 & 0 & 0 \\ 3 & 0 & -1 & -2 \end{bmatrix}^{-1} \begin{bmatrix} 60 \\ 0 \\ -20 \\ 0 \end{bmatrix}$$

Nodal Analysis with Inspection

By inspection, write the nodal voltage equations for the circuit